Storage capacity with a water-activity axis in `sf02.jl`

Implementation and low-resolution tests

StagPP.jl `scripts/sH2O`, branch `new_O_solver`

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1 Summary

The generator now writes the storage capacity at a set of water activities a_k, in a new file, `StagH2Oa.dat`. The old file, `StagH2O.dat`, stays the $a = 1$ slice in the old format, and water-carb reads it unchanged. Each sector applies the activity with its own physics:

- **NAMs**: $c_\varphi(a) = c_\varphi(1)\, a^{n_\varphi}$, with the per-phase exponent of Table 1.
- **DHMS**: $\mu = \mu^\circ_{\mathrm{H_2O}} + RT \ln a_{\mathrm{map}} + RT \ln a$ in `dhms!`. The map is unchanged, and the cell activity is an added term.
- **Upper mantle (ume/mbe)**: MAGEMin with the `aH2O` buffer at $B = a$. In MAGEMin's C source (`pp_min_function.c`) this is $G = G^\circ_{\mathrm{H_2O}} + RT \ln B$, with B clipped to $[10^{-8}, 1 - 10^{-8}]$.

The tests give five results:

1. With a single activity, $a = 1$, and the old fluid set, the new code writes a `StagH2O.dat` byte-identical to the old one (15×15).

2. The first activity run showed that today's $a = 1$ upper-mantle table is not computed with the buffer fluid. At $B \approx 1$, ume's own fluid model `fl` undercuts the buffer and oxidises the rock. As a result, 226 nodes *gained* water as a fell. `fl` is now removed. This changes the $a = 1$ slice in cold upper-mantle peridotite: serpentinite holds 13.9 instead of 12.4–12.8 wt% (Section 4). **Your decision.**

3. After that fix, the capacity decreases with a everywhere except at 1 (15×15, 10 activities), 3 (50×50, 4 activities) and 10 (15×15, 28 activities) MAGEMin local minima. These are capped from $a = 1$ downwards.

4. NAM nodes follow the expected slopes, and the DHMS and serpentine fields recede to lower T as a falls. The worked nodes of `dhms_activity.pdf` are reproduced (phase D → superhydrous B → dry at 18 GPa, phase H out between $a = 0.5$ and 0.25 at 48 GPa).

5. Ten log-spaced activities (the default) give the activity root to better than 0.02 decades for 90 % of the cases and to 0.19 decades at the 99th percentile. Four activities are too few. A production table at 100×100×10 is about 21 MB and should take about 80 min here.

2 What changed in the code

`sfO2_fns.jl`

- `NAM_N`: fugacity exponent per NAM law (Table 1).
- `assemble!(...; a=1.0)`: every NAM law is multiplied by a^n, and opx is split into its En term ($n = 1$) and its Al term ($n = \frac{1}{2}$). Water in hydrous phases comes from the MAGEMin run itself. The MSH water w_{MS} passed to `dhms!` is therefore already scaled.
- `dhms!(...; a=1.0)`: $+RT \ln a$ in μ, and the anhydrous pair member's NAM law scaled by a^n. K&O `hen` maps to the opx En law below 8 GPa and to hpcpx above, as before.

Both default to $a = 1$, so `sH2O_moon.jl` and `sH2O_plot.jl` behave as before.

`sfO2.jl`

- Input `aw` (environment `SFO2_AW`). Default: 1, 0.5, 0.25, 0.1, 0.05, 0.025, 0.01, 0.005, 0.0025, 0.001. It must contain 1.
- `S`, `Shyd` and `Shost` gain a fifth index (activity).
- UM: one ume/mbe run per (a_k, composition) with $B = a_k$. The sb24 hand-over above 5 GPa and the TZ+LM pass run MAGEMin once (the sb24 assemblage is dry and does not depend on a). They redo only `assemble!` and `dhms!` per a_k. The mbe pass of the transition zone runs once per a_k.
- `um_run` also removes `fl` (Section 4).
- Monotone projection: $S(a_k) \leftarrow \min[S(a_k), S(a_{k-1})]$ from $a = 1$ down, with the count of nodes that rose by more than 0.01 %.
- `StagH2Oa.dat`: the layout of `StagfO2.dat` with a in place of R. The header is $n_P\ n_T\ n_a\ n_{EM}$ and the mixes, then the sector bounds, then the a_k line. Blocks run per a_k, then per sector, then basalt, harzburgite and the mixes. `sH2O_maps.jls` also stores `aw`.

Table 1: Fugacity exponent per NAM law (`NAM_N`). $n = 1$ for (2H)Mg or molecular water and $n = \frac{1}{2}$ for single OH (Keppler & Bolfan-Casanova 2006). Constant-capacity laws take 1. All assignments [check].

n	Laws	Note
1	ol, wa, ri, ak, mw (fp), opx En term	(2H)Mg
1	coe/qtz, cf, crst, capv, pl	constant or empirical laws
$\frac{1}{2}$	opx Al term, cpx (di, jd), grt, hpcpx, st, apbo, pv, ppv	single OH, coupled substitution

3 Tests run

Run	Grid (per sector)	Activities	Wall time
`orig`	15×15	1 (old code)	–
`new1`	15×15	1, `fl` kept	–
`g15`	15×15	10 (default)	188 s
`g15f`	15×15	28 (default + 2 geometric points per interval)	≈9 min
`g50`	50×50	1, 0.5, 0.1, 0.01	669 s

All runs used 8 threads on this Mac, `SFO2_DO=sH2O_um,sH2O_deep`, and wrote to `$TMPDIR`. No minimisation failed to converge.

Regression. `orig` against `new1`: `cmp` of the two `StagH2O.dat` reports identical files. The a scaling and the new loops are therefore neutral at $a = 1$.

4 The database fluid at $a \approx 1$

At 2 GPa, 500 K, harzburgite, the ume runs give:

B	Assemblage (wt fractions of the whole system)	Bound H_2O (wt%)
1	atg 0.49, br 0.05, chl 0.03, mgt 0.04, hem 0.01, aug 0.02, `fl` 0.36	12.4
0.9	atg 0.74, br 0.14, chl 0.04, mgt 0.01, aug 0.03, `fl` 0.04, buffer	13.8
0.5	atg 0.77, br 0.15, chl 0.05, aug 0.03, buffer	13.9

At $B = 1$ the buffer phase is absent. The 80 wt% excess water forms ume's fluid model `fl`, and the Fe goes into magnetite and hematite, so brucite is scarce. That is serpentinisation with H_2 going into the fluid. With the pure-H_2O buffer this cannot happen, so the Fe stays in Fe-brucite, which binds water. The bound water therefore rose by up to 2.3 wt% between $a = 1$ and 0.5. That is impossible for a rock open to water at fixed P, T ($\partial n_{H_2O}/\partial\mu \geq 0$). The generator's own comment describes the UM fluid as the pure-H_2O buffer, so the `fl` phase was never intended.

With `fl` removed, $B = 1$ gives the same solid as $B = 0.999$ (the excess water appears as the pure H_2O phase), and the 226 violations drop to 6. Figure 1 shows what this does to the $a = 1$ slice:

- **Sector 1:** 846 of 2250 nodes change, by up to 6.8 wt% at a boundary cell and +0.02 wt% on average.
- **Sectors 2 and 3:** unchanged.
- **Basalt (mbe):** unchanged.

The physical change is in cold serpentinite, 13.9 instead of 12.4–12.8 wt%. The antigorite-out temperature stays the same within the grid spacing. In the production table (old, 100×100) the capacity falls below 2 wt% at 909/939/924/879/833 K at 1/2/3/4/5 GPa. In `g50` (new, $a = 1$) it does so at 898/929/929/898/837 K. The ±3 wt% band in Figure 1 is the coarse 15×15 cell in which that boundary falls.

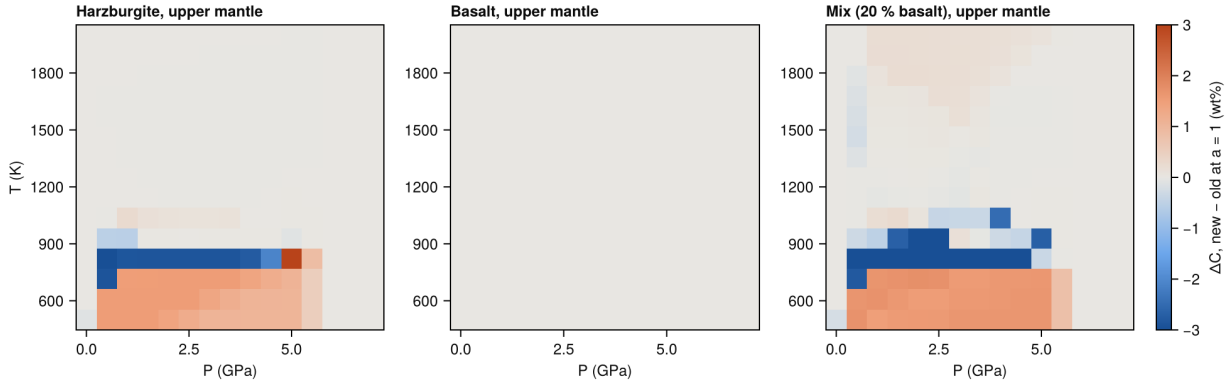


Figure 1: $a = 1$ capacity, new minus old (15×15), upper mantle. Only the removal of `fl` differs.

Decision needed. Keeping `fl` would keep today's $a = 1$ slice, but then the activity axis is not monotone in the serpentine and chlorite fields. Removing it (the current state) gives one consistent fluid at every activity. The cost is that today's $a = 1$ serpentinite gains about 1.1–1.5 wt%, and the oxidation by serpentinisation (magnetite) is lost from the capacity table [check].

5 Monotonicity

Ignoring round-off ($< 10^{-4}$ relative), the remaining rises are isolated MAGEMin local minima. An example is basalt at 3 GPa, 607 K: between $a = 0.7$ and 0.5 MAGEMin switches to a glaucophane-bearing assemblage with more lawsonite, +0.08 wt%. Counts after the fix:

- 1 node (`g15`).
- 3 nodes (`g50`).
- 10 nodes (`g15f`, 28 activities).

They are capped from $a = 1$ downwards, so the $a = 1$ slice is never touched. TOex needs a monotone curve for a unique root, and the cap guarantees it.

6 Capacity curves at single nodes

Figure 2 and Table 2 show the capacity at eight nodes of `g15` over the ten activities.

NAM-only nodes follow the assigned exponents. Olivine at 6.5 GPa gives $C(0.5)/C(1) = 0.503$ ($n = 1$). Stishovite in basalt at 80 GPa gives 0.707 ($n = \frac{1}{2}$). Opx at 3 GPa gives 0.545, between its two terms. Hot ringwoodite at 18.6 GPa gives 0.501.

Hydrous nodes hold their water down to a threshold, then step down:

- **Node A**, harzburgite at 18.6 GPa, 1171 K: phase D (7.90 wt%) down to $a = 0.25$, superhydrous B (4.26) at 0.1–0.05, then dry ringwoodite. This is the sequence of `dhms_activity.pdf`.

- **48.6 GPa, 1171 K**: phase H (11.3 wt%) at $a \geq 0.5$, bridgmanite only from 0.25. This agrees with the analytic threshold 0.56 found at 45 GPa.
- **Serpentinite** (2 GPa, 607 K): antigorite down to 0.1, chlorite down to 0.01, then opx.
- **Basalt** at 3 GPa, 607 K: lawsonite down to 0.1, then talc, then omphacite.

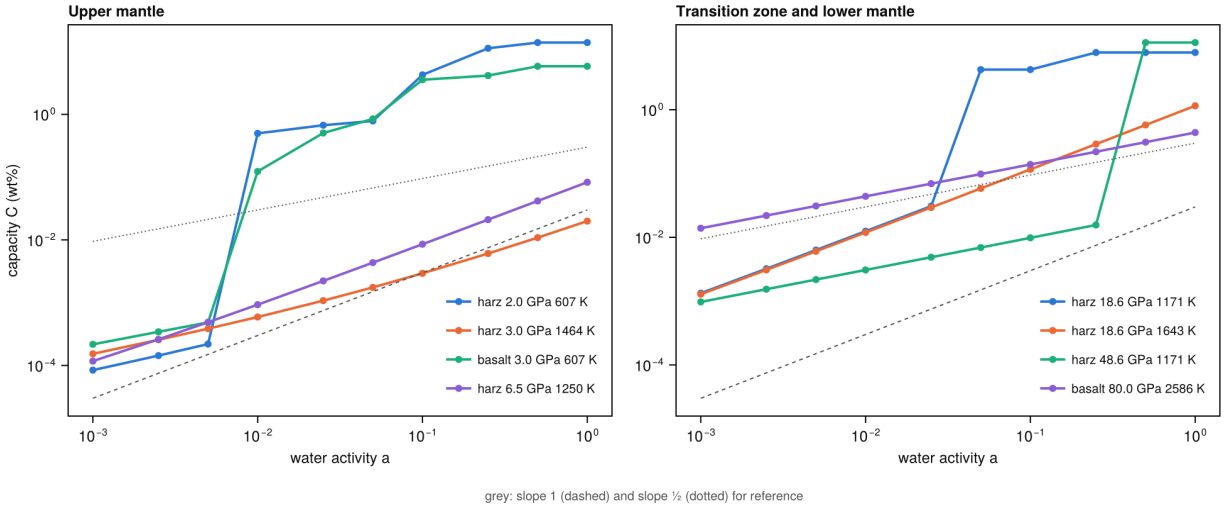


Figure 2: $C(a)$ at eight nodes (`g15`, nearest grid nodes to the targets). Grey lines: slopes 1 and $\frac{1}{2}$.

Table 2: Capacity (wt%) at selected activities, and the main water host.

Node	$a = 1$	0.25	0.1	0.025	0.001	Host as a falls
harz 2.0 GPa 607 K	13.91	11.28	4.25	0.671	8.4e-5	atg → chl → opx
harz 3.0 GPa 1464 K	0.0199	0.0061	0.0029	0.0011	1.5e-4	opx
basalt 3.0 GPa 607 K	5.83	4.15	3.56	0.504	2.2e-4	law → ta → omph
harz 6.5 GPa 1250 K	0.0830	0.0210	0.0085	0.0022	1.2e-4	ol
harz 18.6 GPa 1171 K	7.90	7.90	4.26	0.0308	1.3e-3	phD → shB → ri
harz 18.6 GPa 1643 K	1.153	0.289	0.116	0.0294	1.3e-3	ri
harz 48.6 GPa 1171 K	11.30	0.0156	0.0098	0.0049	9.7e-4	phH → pv
basalt 80 GPa 2586 K	0.439	0.220	0.139	0.0694	0.0139	st

7 Maps at 50×50 with four activities

Figures 3–5 show the capacity, and the share held by hydrous phases, in each sector. Lowering a does two things:

- **NAM capacities** drop uniformly, by 0.5–1 decade per decade of a.
- **Hydrous fields** shrink towards low T. The DHMS boundary of the transition zone moves from about 1450 K at $a = 1$ to about 1250 K at 0.1 and about 1100 K at 0.01. Phase H below 60 GPa recedes in the same way. Antigorite and chlorite in the upper mantle shrink to below about 750 K by $a = 0.01$.

Figure 6 condenses this into $n_{\mathrm{eff}} = \ln[C(0.1)/C(1)]/\ln 0.1$:

- $n_{\mathrm{eff}} \approx \frac{1}{2}$ or 1 where NAMs hold the water.

- $n_{\mathrm{eff}} > 1$ in bands where a hydrous phase breaks down between the two activities.
- $n_{\mathrm{eff}} \approx 0$ where the hydrous phase survives to $a = 0.1$.

Composition	Sector	$n_{\mathrm{eff}} \leq 0.55$	0.55–1.05	> 1.05
Harzburgite	upper mantle	0.07	0.67	0.25
Harzburgite	transition zone	0.37	0.55	0.08
Harzburgite	lower mantle	0.92	0.05	0.03
Basalt	upper mantle	0.68	0.03	0.29
Basalt	transition zone	0.86	0.11	0.03
Basalt	lower mantle	0.95	0.02	0.03

The shares show why one exponent per rock type would not do. Harzburgite is mostly $n = 1$ in the upper mantle and mostly $\frac{1}{2}$ in the lower mantle, and basalt is mostly $\frac{1}{2}$ everywhere.

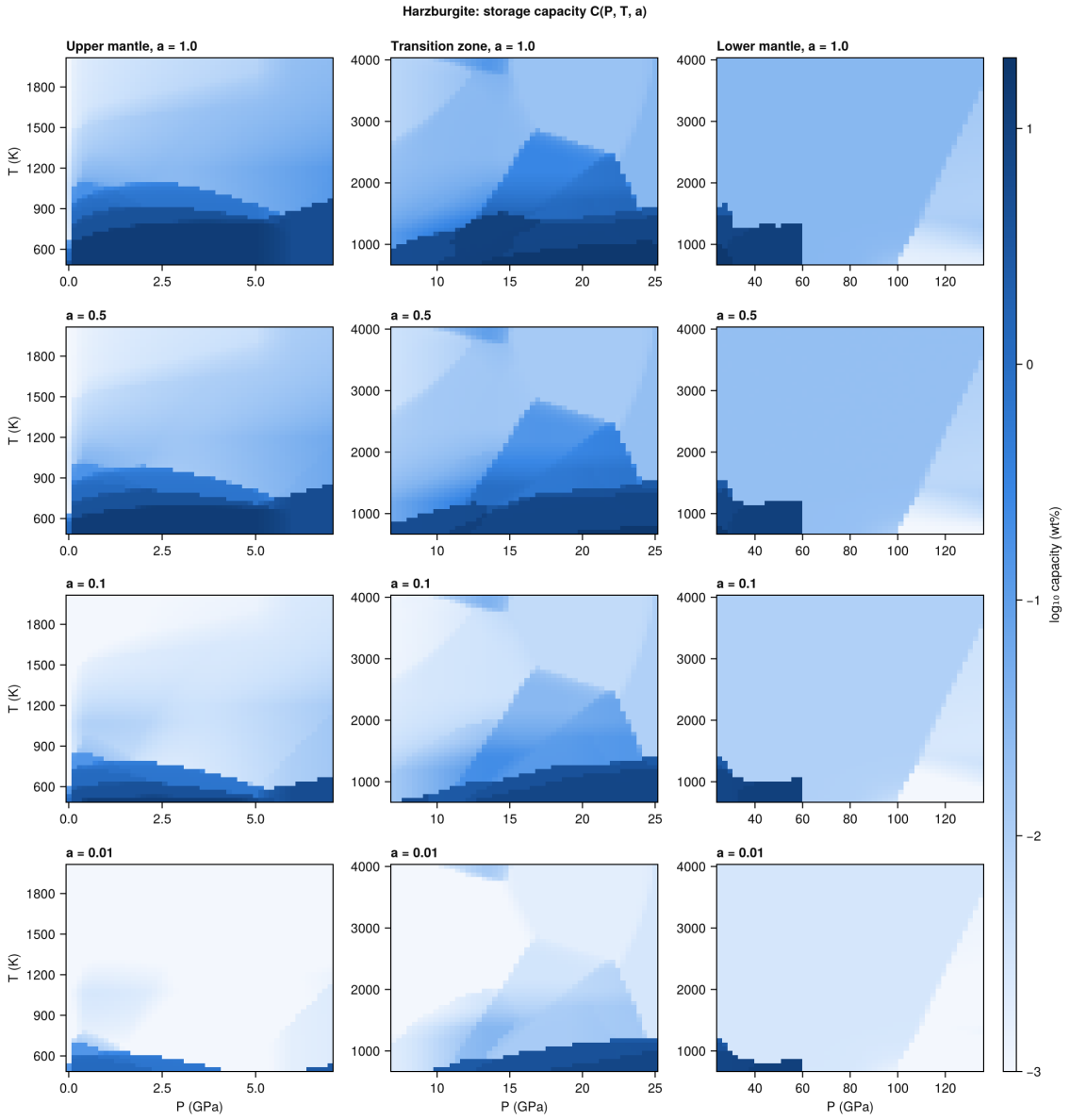


Figure 3: Harzburgite, $\log_{10} C$ at $a = 1,\ 0.5,\ 0.1,\ 0.01$ (g50).

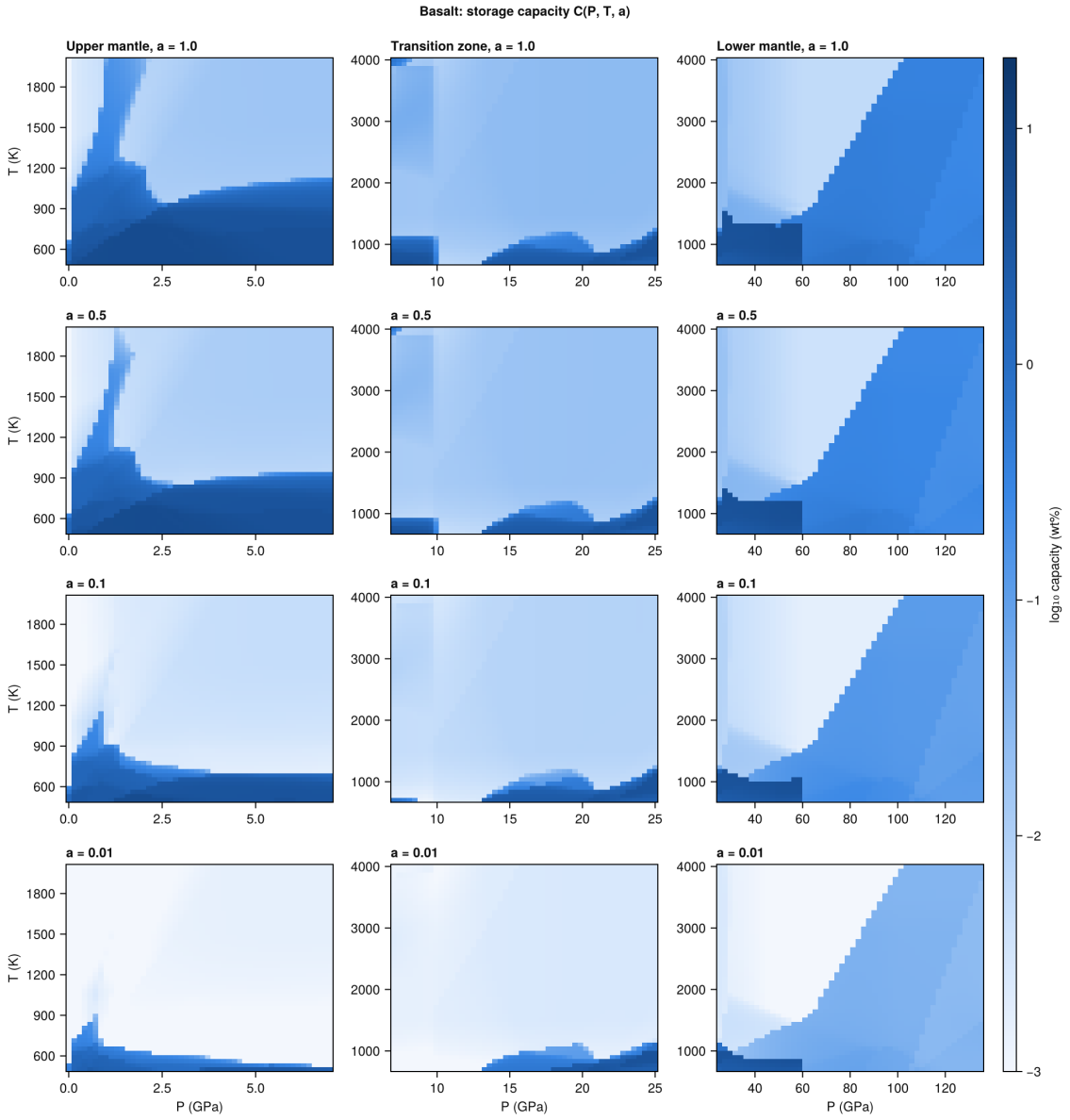


Figure 4: Basalt, $\log_{10} C$ at $a = 1$, 0.5, 0.1, 0.01 (g50).

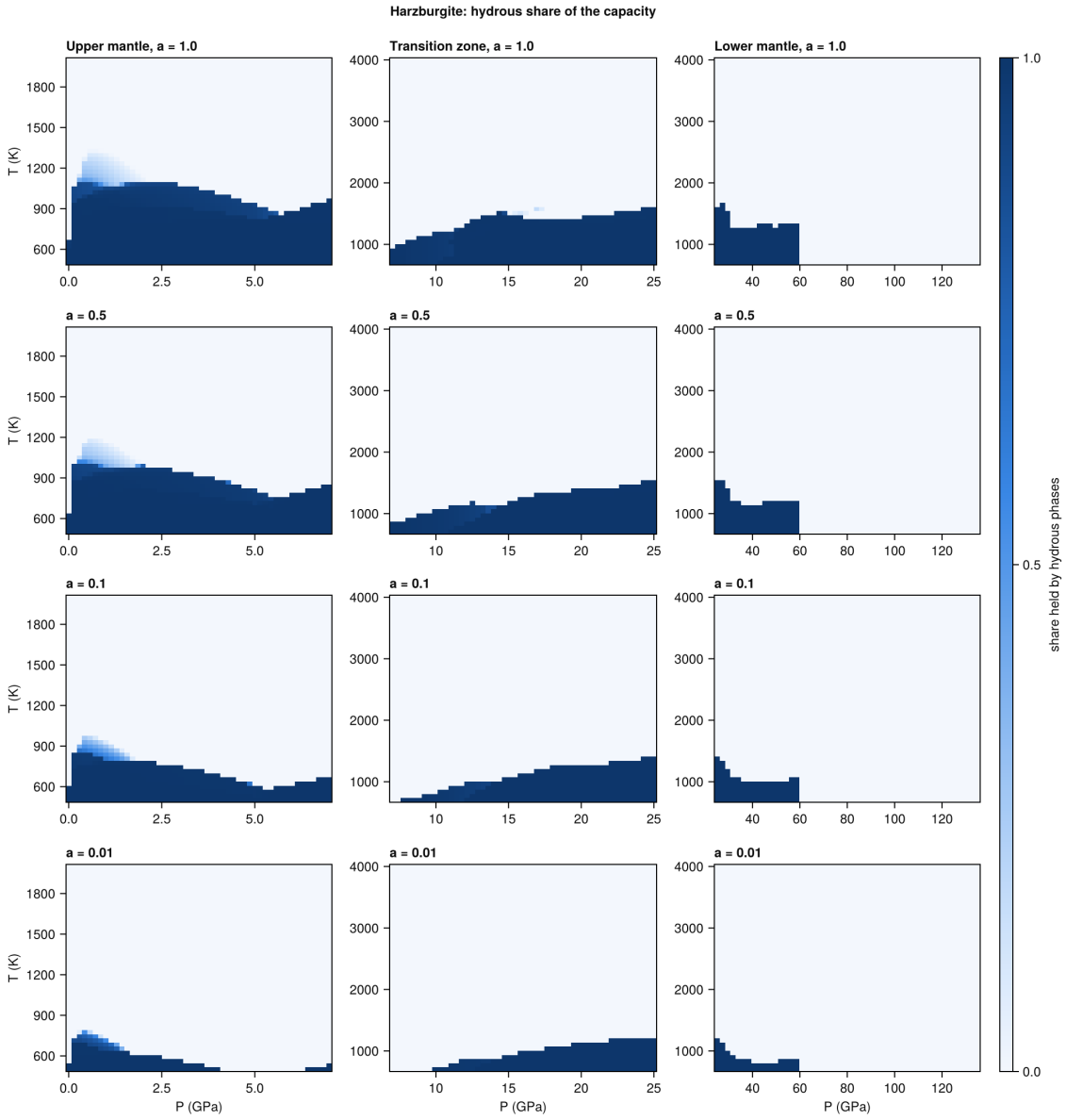


Figure 5: Harzburgite, share of the capacity held by hydrous phases (g50).

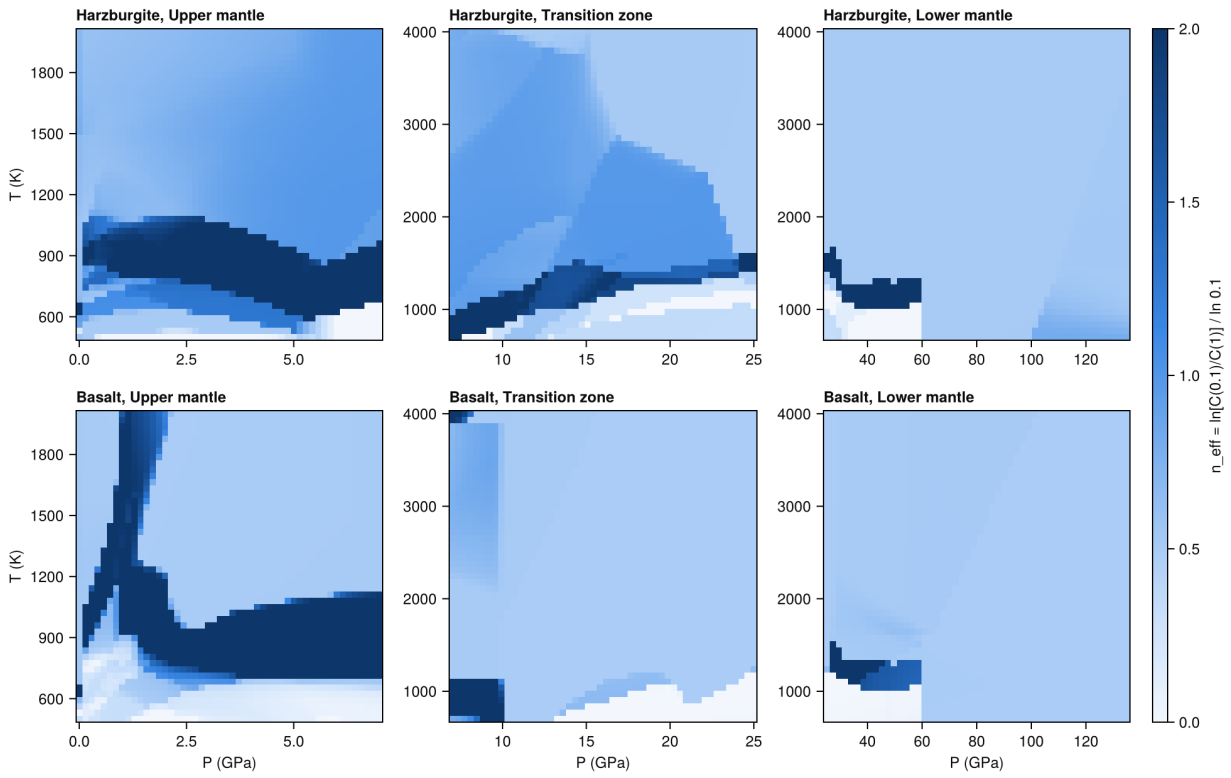


Figure 6: Effective exponent between $a = 1$ and $a = 0.1$ (g50).

8 How many activities the table needs

TOex would interpolate the table linearly in $(\ln a, \ln C)$, with the last slope below the lowest node. That reproduces a pure power law exactly, so only the hydrous steps are smeared. The 28-activity run (`g15f`) serves as the reference. Three sub-grids are interpolated to the activities they leave out. Two errors are measured. The first is the capacity at that activity. The second is the one that matters for the solve: the activity root that the interpolated curve returns for the true capacity (Figure 7).

	$\lvert C_{\rm tab}/C - 1\rvert$		$\lvert\Delta \log_{10} a\rvert$ (decades)			
Grid	median	p90	median	p90	p99	share > 0.1
4 nodes (1, 0.5, 0.1, 0.01)	1.5e-3	0.26	1.5e-3	0.14	1.0	13.2 %
10 nodes (default)	1.3e-4	5.2e-3	1.6e-4	0.018	0.19	3.6 %
19 nodes (default + midpoints)	6.0e-5	2.2e-3	7.4e-5	5.8e-3	0.11	1.9 %

The tail comes from the dehydration steps. Between two nodes that straddle a step, the table spreads the step over the whole interval. The largest capacity error, 11 wt%, is therefore the same for every grid. The activity error is bounded by the node spacing. With ten nodes, 0.3–0.4 decades apart, a step is placed to within about 0.2 decades in 99 % of cases. That is smaller than the uncertainty of the DHMS map itself (the K&O fitted activities alone span 0.012–0.85). Doubling to 19 nodes halves the tail at twice the cost. **Recommendation: keep the 10-node default.** Below $a = 10^{-3}$ the extrapolation uses the last slope, which is the NAM power law in every node tested.

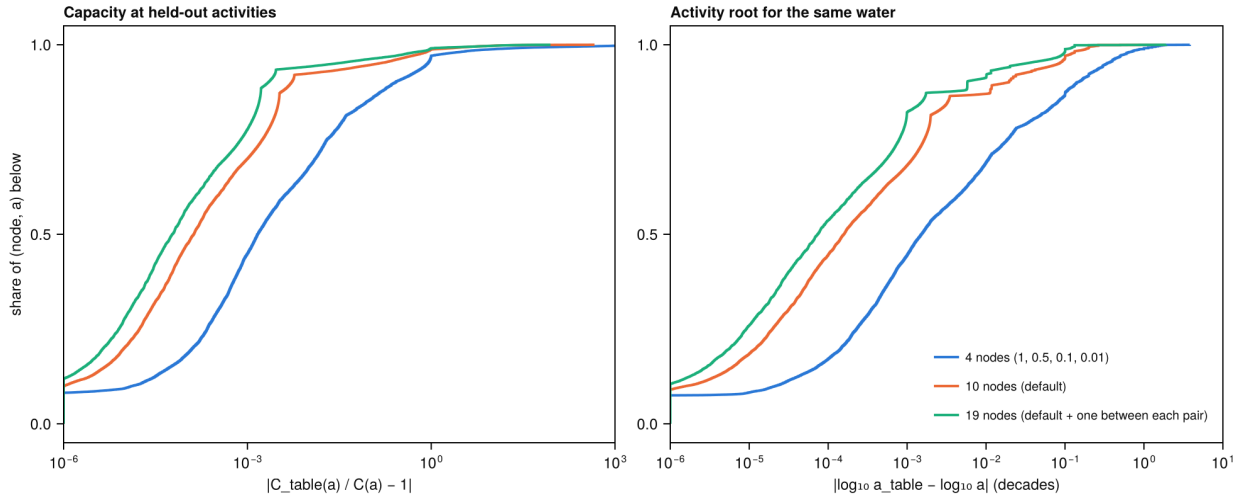


Figure 7: Cumulative distributions of the two errors over every (node, held-out activity) of `g15f`.

9 Cost of the production table

- **Size.** `StagH2Oa.dat` at 50×50×4 is 2.1 MB. At 100×100×10 it is about 21 MB, a quarter of `StagfO2.dat` (85 MB). `StagH2O.dat` keeps its 2.1 MB.
- **Time.** In the existing production run, one ume/mbe composition takes 32–46 s at 100×100 (`out/run_sH2O_pmbe065.log`), and the TZ+LM sb24 pass 446 s. With ten activities the UM costs about $10 \times 10 \times 40$ s $\approx$ 67 min. The deep pass still runs MAGEMin once. Each extra activity adds only `assemble!` and `dhms!`, a few seconds. In total about 80 min on 8 threads here [check].

10 Open points before the production run

1. **`fl` in the UM runs**: keep removed (consistent axis, $a = 1$ serpentinite +1.1–1.5 wt%), or restore it (today's $a = 1$ slice, non-monotone axis)? Section 4.

2. **Exponents** of Table 1: each assignment should be checked against the incorporation mechanism in its source [check]. hyWad and hyRin stay on the wa and ri laws ($n = 1$).

3. **The cap** keeps $a = 1$ and lowers the lower slices. The alternative, raising from the bottom, would change the $a = 1$ slice. Both are minimiser repairs.

4. `sH2O_math.md` needs the activity axis, the exponents, the reading of a_{map} as an effective correction (S.7), and the removal of `fl`.

5. **watercarb, `StagPP.jl/src` and the MAGEMin stand-alone copy**: the `StagH2Oa.dat` reader, the per-tracer vector lookup, and the solid curve in TOex. Not started, waiting for your go.

6. **Test scripts** are in `$TMPDIR/aw_test` (cleared periodically). If they are worth keeping, they should be moved into the repository.